



CHALLENGE 05 / 20

THE SILENT RANSOMWARE FACTORY

PRE-DETONATION ATTACKER PRODUCTION-PROCESS RECONSTRUCTION

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A group with eleven engineering plants and 7,400 endpoints and servers had 2,300 systems encrypted in 41 minutes at 02:10. On every host the payload was different: structure, strings, packing, execution order and ransom note text. No two hashes matched. The attackers had been inside for six to ten weeks and destroyed backups first. Some reachable hosts were not encrypted.

Recovery depends on facts nobody has: what generated these payloads, where they were produced and from what, which hosts served as build or staging points, whether footholds remain, and whether unencrypted hosts were prepared for a second wave. What exists is a population of unique artefacts and partial network and host traces.

Conventional signature-based response is useless. Each payload looks like a separate piece of malware.

2 | PROBLEM STATEMENT

Develop a **Payload Origin and Production Reconstruction System** that, from thousands of individually unique malicious artefacts and partial environmental traces, infers the shared generator, its configuration and the production pipeline, identifies production and staging systems, and reconstructs the pre-detonation timeline.

The system must also predict which hosts that were not detonated were prepared for a later wave.

Hard constraints

- Variation between artefacts is deliberate and large; similarity measures that work on ordinary malware fail.
- Environmental evidence is partial, and some of it was destroyed by the attacker.
- The system must separate several operators or toolchains if more than one was involved.
- Teams must use benign synthetic stand-ins for artefacts. No functional malware may be created or used.
- Timeline and second-wave predictions must come with bounds that are shown to be valid in the team's own tests.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Polymorphic Population Analysis	Ability to analyse large populations of deliberately varied artefacts.
02	Generator Inference	Ability to infer the shared generator and its configuration.
03	Multi-Operator Separation	Ability to separate more than one toolchain or operator.
04	Production Site Identification	Ability to identify build and staging systems from indirect traces.
05	Pre-Detonation Timeline	Ability to reconstruct and bound the preparation period.
06	Second-Wave Prediction	Ability to predict which undetonated hosts were prepared.
07	Destroyed-Evidence Reasoning	Ability to reason about evidence the attacker removed.
08	Inference Traceability	Ability to link each inference to the observations behind it.

The system must clearly distinguish between:

OBSERVED	INFERRED	ESTIMATED	HYPOTHESIS	UNKNOWN
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A dwell-time or origin claim that cannot be supported must not be presented as established.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated plant estate with benign synthetic artefacts
02	Large artefact population with deliberate variation
03	Recovery of the generator and its configuration
04	Identification of production and staging systems
05	Separation of two simulated toolchains
06	Pre-detonation timeline with bounds
07	Second-wave host prediction

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.